Towards Effective Application of Life Cycle Assessment in Manufacturing: Challenges and Recommended Strategies

Abstract
This study examines the challenges and enablers of implementing Life Cycle Assessment (LCA) in manufacturing. Using a systematic literature review (2020–2025) and expert validation, it identifies technical, methodological, and organizational barriers and develops typologies and frameworks to support practical decision-making. The findings aim to bridge the gap between theory and practice, providing actionable guidance for sustainable production integration.

Keywords: Life Cycle Assessment, Sustainable Manufacturing, Systematic Literature Review, LCA Implementation, Industrial Practice, Decision-Support Frameworks

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1. Introduction

Over the past half-century, Environmental Life Cycle Assessment (LCA) has evolved into a key methodology for evaluating the environmental impacts of products and processes. The earliest studies, emerging in the late 1960s and early 1970s (Sundström, 1971), reflected growing public concern over resource efficiency, pollution control, and waste management (Guinée, 2016). Between the 1970s and 1990s, LCA development was marked by diverse approaches and inconsistent methods due to the absence of international coordination, resulting in highly variable outcomes even for similar studies (Guinée et al., 1993).

The early twenty-first century brought renewed attention to LCA, supported by influential publications and global initiatives such as life cycle–based carbon footprint standards (Guinée et al., 2002; Baumann & Tillman, 2004; EC, 2010; Curran, 2012; Klöpffer & Grahl, 2014). During this period, LCA gained prominence as a strategic tool for environmental policy and industrial sustainability. However, as the field expanded, methodological refinements also led to increased divergence in practices, with researchers proposing varying approaches to system boundaries, allocation rules, and data treatment. Emerging variants such as dynamic, spatially differentiated, and hybrid LCAs further diversified the field, often complicating result comparability and standardization (Guinée, 2016). The integration of Life Cycle Costing (LCC) and Social Life Cycle Assessment (SLCA) broadened the traditional environmental scope of LCA into a Life Cycle Sustainability Assessment (LCSA) framework, reflecting the three-pillar model of sustainability and the principles of Industrial Ecology (IE).

On the other hand, a profound transformation of industrial systems is anticipated worldwide as industries move toward more sustainable modes of production. Achieving large-scale sustainability transitions requires not only changes in production practices but also significant upgrading and redesign of existing equipment and manufacturing systems. Legacy production assets must be modernized and optimized, while new systems need to be designed for enhanced resource efficiency and adaptability to meet the growing global challenges of environmental impact, resource scarcity, and economic competitiveness (Birkie et al., 2025). Although numerous researchers have proposed various definitions and methodological frameworks for LCSA in recent years, industrial practitioners continue to face uncertainty regarding its precise meaning, methodological application, and appropriate contexts for use. This ambiguity highlights a gap between theoretical development and practical implementation of LCSA. Consequently, an important question arises: how is LCSA being applied in industrial practice?

Given this context, the theoretical foundations of LCA and its extension into LCSA have advanced considerably, their effective implementation in manufacturing systems remains limited and inconsistent. The transition toward sustainable production demands not only methodological refinement but also practical strategies that enable industries to embed LCA within real-world decision-making processes. Therefore, this research aims to contribute to the ongoing discourse by identifying the key challenges hindering effective LCA application in manufacturing and by suggesting strategies that can support a more systematic, transparent, and impactful use of LCA as a decision-support tool for sustainable industrial transformation.

2. Preliminary Literature Review

2.1. Applications and Methodologies of Life Cycle Assessment in Manufacturing

Over recent decades, Life Cycle Assessment (LCA) has evolved from a theoretical framework into a practical and data-driven methodology for evaluating environmental impacts within manufacturing contexts. The growing emphasis on sustainability and digital transformation has driven the adaptation of LCA methodologies to suit complex industrial systems. Piron et al. (2024) conducted a systematic review examining how Industry 4.0 technologies such as the Internet of Things (IoT), cyber-physical systems (CPS), and cloud-based data infrastructures enhance LCA implementation by improving data collection, traceability, and accuracy within the Life Cycle Inventory (LCI) phase. These technologies allow for real-time monitoring of energy use, emissions, and resource flows, enabling more dynamic and responsive LCAs that reflect actual operational conditions.

Similarly, Kiemel et al. (2022) investigated methodological approaches to simplify LCA applications in industrial settings. Their study categorized simplification methods into five main strategies: parametric, modular, automation, aggregation, and screening. Parametric models, for instance, allow users to adjust input variables to rapidly assess multiple design configurations, while modular LCAs break down complex systems into smaller, manageable sub-processes. Although these strategies enhance scalability and accessibility, data collection for complex products and multi-tier supply chains remains a major challenge, often resulting in incomplete or inconsistent inventories.

In an effort to make LCA more accessible to non-experts, Wang et al. (2022) introduced a knowledge-enriched LCA framework that leverages knowledge graphs to support inventory data recommendations and semantic reasoning. By connecting domain knowledge and contextual data sources, such systems facilitate automated data retrieval and model generation, potentially reducing the expertise barrier that often hinders industrial LCA adoption. Collectively, these developments signal a clear trend toward the integration of digital tools, artificial intelligence, and knowledge-based systems in enhancing LCA's methodological robustness and practical usability in manufacturing.

2.2. Key Challenges, Barriers, and Limitations

Despite these technological and methodological advancements, the widespread adoption of LCA in manufacturing continues to face several systemic challenges. Kaswan and Rathi (2021) identified critical barriers including insufficient management commitment, limited employee training, and ineffective data management across supply chains. These organizational constraints often result in fragmented LCA initiatives that fail to inform broader strategic decisions. Moreover, inadequate understanding of LCA principles among practitioners can lead to misinterpretation of results and suboptimal decision-making.

In the context of additive manufacturing, Naser et al. (2023) reported that high data intensity and low automation significantly limit the scalability of LCA. They noted that LCA is rarely integrated into early design stages, despite its potential to influence material selection, process planning, and product architecture. Similarly, Chen et al. (2024) found that in the pharmaceutical industry, LCA applications are constrained by narrow system boundaries, inconsistent data presentation, and the lack of comprehensive databases. These limitations hinder cross-comparison among studies and reduce the credibility of results in regulatory and policy contexts.

Together, these findings illustrate that while LCA methodologies have matured, operationalization barriers persist. Complex manufacturing environments, characterized by heterogeneous data systems, long supply chains, and varying regulatory contexts, amplify these difficulties. As such, effective LCA implementation demands not only technical tools but also institutional commitment, workforce competence, and cross-sector collaboration to standardize practices.

2.3. Conceptual Frameworks, Guidelines, and Recommendations

In response to the aforementioned challenges, several conceptual frameworks and guidelines have been developed to improve LCA implementation and reliability in manufacturing.

Schneider et al. (2024) proposed an integrated LCA–FMEA framework, combining environmental assessment with risk analysis to evaluate both sustainability performance and potential system failures. This hybrid method allows for the simultaneous consideration of environmental, technical, and safety-related factors thereby enhancing the strategic value of LCA in production system design.

Valdivia et al. (2021) advanced the theoretical underpinnings of Life Cycle Sustainability Assessment (LCSA) by introducing ten guiding principles emphasizing completeness, transparency, consistency, and alignment with ISO 14040 standards. These principles advocate a holistic and multi-dimensional assessment approach that integrates environmental, economic, and social perspectives. Meanwhile, Cucchi et al. (2022) validated the application of Organizational Life Cycle Assessment (O-LCA) in industrial contexts through the use of digital technologies such as IoT and cloud computing. Their findings demonstrate that digital integration enables dynamic, near-real-time monitoring of environmental performance, thereby supporting continuous improvement in manufacturing operations.

These frameworks collectively underscore the need for structured methodologies that not only ensure methodological rigor but also facilitate decision-making. The convergence of LCA with risk assessment, digital analytics, and organizational management practices represents a promising direction for advancing sustainability-oriented production strategies.

3. Research Gap and Research Questions

Although Life Cycle Assessment (LCA) has evolved substantially over the past five decades from fragmented environmental assessments to a standardized, ISO-based methodology the literature reveals persistent gaps that hinder its effective implementation in manufacturing contexts. Despite extensive theoretical development, practical adoption remains inconsistent and limited, particularly in complex and data-intensive industrial environments.

First, there is a clear gap between methodological advancement and practical implementation. While recent research demonstrates the potential of digital tools such as IoT, cyber-physical systems, and knowledge graphs to enhance data collection and automation (Piron et al., 2024; Wang et al., 2022), these technologies are not yet widely integrated into everyday manufacturing practices. Many industries continue to rely on static and manual approaches to data gathering, leading to incomplete or outdated Life Cycle Inventories (LCIs). This lack of digital integration limits the operational value of LCA as a real-time decision-support tool.

Second, organizational and systemic barriers persist despite growing awareness of sustainability imperatives. Studies (Kaswan & Rathi, 2021; Naser et al., 2023) consistently identify obstacles such as insufficient managerial support, lack of employee training, and fragmented supply chain data as critical impediments. These barriers underscore that LCA challenges are not merely technical, but also institutional and behavioral. Consequently, many organizations struggle to embed LCA outcomes into strategic and operational decision-making processes.

Third, methodological inconsistencies continue to undermine comparability and reliability. Researchers such as Chen et al. (2024) have noted that unclear system boundaries, variable data quality, and the absence of comprehensive databases lead to divergent results, even when similar systems are analyzed. Despite the availability of international standards, LCA practice remains context-dependent and often customized to specific project needs, preventing generalization and large-scale benchmarking across industries.

Fourth, existing frameworks have yet to fully realize the integration of LCA within broader sustainability assessments. While initiatives such as Life Cycle Sustainability Assessment (LCSA), Organizational LCA (O-LCA), and hybrid LCA–FMEA frameworks (Valdivia et al., 2021; Schneider et al., 2024; Cucchi et al., 2022) offer promising directions, their practical application in manufacturing systems remains underexplored. The literature reveals limited empirical validation of how such frameworks can be operationalized to support production planning, system redesign, or policy compliance in real industrial settings.

Finally, there is a lack of harmonized and user-friendly tools to democratize LCA adoption, especially for small and medium-sized enterprises (SMEs). Current tools often require advanced expertise and substantial resources, making LCA inaccessible to non-specialists. The limited automation and absence of intuitive interfaces hinder broader engagement and prevent LCA from being used as a proactive management tool rather than a retrospective evaluation method.

Taken together, these gaps highlight that while the theoretical and methodological foundations of LCA are well-established, its translation into actionable practice within manufacturing remains incomplete. The need for clearer conceptual guidance, standardized digital integration, and organizational enablers is evident. Therefore, this study seeks to address two central questions:

- **RQ1:** What are the key challenges in applying LCA effectively in manufacturing industries?
- **RQ2:** What are the recommended strategies that can be implemented by the manufacturing industry to overcome the challenges of implementing LCA?

By systematically investigating these questions, the study aims to bridge the divide between theory and practice, contributing to the development of structured, transparent, and practical strategies that facilitate the effective implementation of LCA within sustainable manufacturing systems.

4. Methodology

4.1. Systematic Literature Review

This study adopts a Systematic Literature Review (SLR) methodology to investigate the challenges and boundaries of implementing Life Cycle Assessment (LCA) in production systems. The SLR approach was chosen for its exploratory and analytical nature, which enables the systematic synthesis of existing research to identify both theoretical and practical knowledge gaps. By employing a structured and evidence-based research design, this study aims to establish a robust foundation for advancing academic understanding and supporting theoretical development on LCA integration in manufacturing. This aligns with established guidelines for conducting systematic reviews, which emphasize transparency, replicability, and minimized bias through predefined procedures for searching, identifying, appraising, evaluating, and summarizing studies.

The purpose of this methodology is to comprehensively address the central research questions of this proposal:

- **RQ1:** What are the key challenges in applying LCA effectively in manufacturing industries?
- **RQ2:** What are the recommended strategies that can be implemented by the manufacturing industry to overcome the challenges of implementing LCA?

4.1.1. Research Design

The SLR process was designed in accordance with recognized scientific principles, ensuring methodological rigor and traceability. It followed three major phases: (1) planning, (2) execution, and (3) synthesis and validation. In the planning phase, research objectives were defined, search parameters were established, and review criteria were agreed upon by all researchers. This systematic approach allows for objective and comprehensive coverage of secondary data sources such as scientific journals, industrial reports, and sustainability assessments.

The SLR is particularly suited to this study's aims because it draws upon validated secondary data to build a theoretical and empirical foundation for understanding LCA's role in sustainable production. It enables an evidence-based evaluation of both the limitations of current applications and the potential strategies that can enhance decision-making in industrial contexts.

4.1.2. Search and Selection Process

The literature search will be conducted across leading academic databases, specifically KTH Library databases that are connected to ScienceDirect, Scopus, and Google Scholar that will ensure broad coverage of relevant and peer-reviewed sources. Boolean search combinations were used to refine results and capture the intersection of sustainability and production systems, such as:

> ("Life Cycle Assessment" OR "LCA" OR "Life Cycle Thinking") AND ("production system" OR "manufacturing process") AND ("sustainability" OR "decision-making").

The inclusion criteria required that studies:

1. Explicitly examine LCA applications within manufacturing or production systems;
2. Address sustainability, process optimization, or decision-making integration;
3. Be peer-reviewed and published between 2020 and 2025; and
4. Be written in English with accessible full-text content.

Papers that lacked empirical data, are not related to industrial contexts, or do not present methodological insight will be excluded. 675 journals will go through the screening process based on the inclusion and exclusion criteria.

To ensure balanced division of workload and analytical depth, the papers were distributed among three researchers. Each researcher will conduct an in-depth review of their assigned papers, documenting findings in a shared database and categorizing results based on recurring themes such as LCA methodologies, implementation barriers, decision-support integration, and sustainability frameworks.

4.1.3. Data Extraction dan Coding

Data extraction follows a structured coding process designed to ensure comparability across studies. Each paper will be reviewed using a consistent analytical framework that captured the following dimensions:

- Context and industrial and manufacturing sector (e.g., automotive, process manufacturing, additive manufacturing)
- LCA Type (product-oriented, process-oriented, or hybrid)
- Challenges Identified (technical, methodological, organizational, or systemic)
- Sustainability Aspects Addressed (environmental, social, or economic)
- Decision-making Models or Frameworks Proposed
- Key Findings and Lessons Learned

The coding framework was organized around two primary research questions.

For RQ1, the review identifies recurring challenges and limitations that hinder the effective application of LCA in manufacturing industries, including technical and methodological issues, such as poor data quality, inconsistent system boundaries, lack of standardized inventory databases, and high resource demands during data collection and analysis. Organizational challenges such as limited expertise, insufficient management engagement, and financial constraints will be also highlighted across multiple studies.

For RQ2, the review focuses on identifying recommended strategies and best practices that can be implemented by the manufacturing industry to overcome these challenges. The analysis will emphasize approaches such as the automation and digitalization of LCA workflows (Schneider et al., 2023), integration of machine learning and simulation-based models for predictive LCA (Naser et al., 2023), hybrid environmental-economic assessment frameworks combining LCA and Life Cycle Costing (Räikkönen et al., 2023), and organizational measures like capacity building, data standardization, and cross-departmental collaboration. Together, these strategies represent practical pathways for improving the scalability, reliability, and decision-making relevance of LCA in modern manufacturing environments.

4.1.4. Data Synthesis and Validation

A thematic synthesis approach will be employed to interpret the extracted data and identify patterns, interrelations, and research trends. The synthesis results will be organized into conceptual categories corresponding to the research questions. To ensure internal validity and minimize researcher bias, the findings from each subset of papers will be cross-reviewed among the three team members through iterative discussion sessions. The triangulated results will then be compared with data from the forthcoming manufacturing case study, serving as an external validation mechanism for the insights derived from the literature.

4.2. Case Study

4.2.1. Empirical Validation through Expert Interviews

Following the completion of the Systematic Literature Review (SLR), an interview-based validation phase will be conducted to triangulate and enrich the findings. While the SLR provides theoretical insight into challenges and frameworks for LCA integration, the interviews will help capture practitioners' experiences, context-specific barriers, and organizational realities that may not be evident in published studies. This dual-method approach strengthens external validity and enables the proposal to bridge the gap between theory and industrial practice, as recommended by Birkie et al. (2025).

The empirical phase adopts a qualitative, semi-structured interview design, inspired by the directed qualitative content analysis method used by Birkie et al. (2025). Interviews will target industry professionals and sustainability experts directly involved in manufacturing system design, environmental management, or LCA application. The selection criteria will be:

- Professionals with hands-on experience in LCA, sustainability reporting, or production system development.

- Representation from different manufacturing sectors (e.g., automotive, electronics, process industry).
- Preference for firms actively working with sustainability tools (e.g., ISO 14040, carbon footprinting, or digital twins).

A total of 8–10 experts are expected to participate, ensuring diversity of perspectives while maintaining analytical depth and an interview guide will be developed based on the SLR findings and structured around the three research questions (RQ1 and RQ2).

4.2.2. Data Analysis and Coding

A deductive–inductive thematic analysis will be used:

- Deductive coding: Based on predefined themes from the SLR (e.g., methodological, organizational, and data-related challenges).
- Inductive coding: Capture new themes emerging from the interview data.

Coding will follow a shared template across all interviews, including:

- Context (sector, company type)
- LCA application type (product/process/system-level)
- Identified challenges
- Suggested improvements or enablers
- Alignment with conceptual recommendations from literature

Each transcript will be coded independently by two researchers to enhance inter-rater reliability, followed by consensus discussions.

5. Theoretical Framework

This research is based on three interrelated foundations that create the Life Cycle Assessment (LCA) framework which are Life Cycle Thinking (LCT), Sustainability (Triple Bottom Line), and Decision-making and Project Management Integration Models (Räikkönen et al., 2023). All of these frameworks collectively support the analysis of LCA implementation within the production system and form the conceptual foundation of the study.

Life Cycle Thinking (LCT) is one of the theoretical cores of LCA, emphasizing a cradle-to-grave perspective of products and processes. It encourages industries to assess the environmental implications of each stage of production, from raw material extraction to end-of-life disposal. In manufacturing, LCT enables companies to identify hidden environmental hotspots, revealing trade-offs between process optimization, material efficiency, and waste reduction. Räikkönen et al. (2023) argue that embedding LCT in production design supports more systemic and preventive sustainability strategies, rather than reactive end-of-pipe solutions. Hence, LCT functions not merely as an analytical lens but as a guiding mindset that connects engineering decisions to environmental responsibility.

Another core of LCA is the sustainability concept through the Triple Bottom Line (TBL) that expands the assessment beyond environmental considerations to include economic and social dimensions. This view includes "People, Planet, and Profit" which frames production systems as socio-technical ecosystems that must balance resource efficiency, financial viability, and social welfare. In LCA applications, the environmental pillar quantifies emissions and material flows, the economic pillar integrates Life Cycle Costing (LCC), and the social pillar evaluates labor conditions and community impacts (Haddad et al., 2023). Integrating TBL into LCA consequently ensures that sustainability is not reduced to ecological metrics but is approached holistically.

Finally, Decision-making and Project Management Integration Models provide the structural mechanism for applying LCA insights in practice. Manufacturing decision-makers often face challenges in translating LCA outcomes into actionable strategies due to data complexity and the lack of integration with management systems (Schneider et al., 2023). Integrative models such as stage-gate processes, continuous improvement loops, and automated LCA dashboards can be embedded in environmental metrics within operational planning, enabling sustainability-oriented decisions at the managerial level. By linking LCA with project management tools, organizations can move from static assessments to dynamic, data-driven sustainability strategies.

Together, these frameworks form a coherent structure that connects production practices with sustainability assessment and decision-making. This study conceptualizes the relationship through the following model:

Manufacturing Systems → Life Cycle Assessment (LCA) → Identified Challenges & Boundaries → Conceptual Strategies for Sustainable Production Integration

This model represents the analytical pathway guiding this research: understanding manufacturing processes, assessing them through LCA, identifying implementation challenges, and ultimately proposing a conceptual framework to bridge theory and practice.

6. Expected Result and Discussion

Based on the systematic literature review (SLR) and planned expert interviews, this study is expected to produce a set of structured outputs that support the effective application of Life Cycle Assessment (LCA) in manufacturing. The outputs are justified by prior research demonstrating that SLRs, particularly when combined with expert validation, consistently produce actionable frameworks, typologies, and research insights (Kiemel et al., 2022; Gómez-Garza et al., 2024; Lee et al., 2023).

The key expected outputs are summarized in the following table:

Expected Output	Description	Justification
Structured list of challenges/barriers in LCA application	Technical, methodological, organizational, and systemic challenges affecting LCA adoption in manufacturing.	Previous SLRs identified recurring barriers and categorized them effectively (Kiemel et al., 2022; Gómez-Garza et al., 2024).
Conceptual recommendations / frameworks	Strategies or models to integrate LCA into production decision-making, including digital integration, knowledge frameworks, and procedural guidelines.	SLRs combined with expert interviews validate and refine recommendations for practical applicability (Lee et al., 2023; Gulotta et al., 2023).
Categorization / typology of challenges and recommendations	Classification of challenges into technical, methodological, and organizational categories; recommendations into strategic, operational, and digital enablers.	Multi-level classifications are commonly produced in LCA SLRs to facilitate knowledge transfer and implementation (Kiemel et al., 2022; Lee et al., 2023; Gulotta et al., 2023).
Identification of research gaps and future directions	Highlighting areas where LCA is underdeveloped, including integration of socio-economic dimensions, digital tools, or methodological harmonization.	SLRs systematically map knowledge gaps and guide future research in LCA and sustainability studies (Mesa et al., 2021; Kumar et al., 2025).
Validation through expert input	Comparing SLR findings with practitioners' experiences to prioritize recommendations and assess practical feasibility.	Expert interviews or workshops are effective for contextualizing and validating SLR findings, enhancing robustness and applicability (Lee et al., 2023).

Table 1: Summary of expected results and strategies

7. Sustainability and Ethical Aspects

Life Cycle Assessment (LCA) evaluates environmental impacts by quantifying emissions, resource use, and ecological footprints, guiding strategies for pollution reduction and resource conservation (Tamym et al., 2023; Chang et al., 2017). Complementarily, Social LCA (S-LCA) and Life Cycle Costing (LCC) assess social equity, labor conditions, stakeholder well-being, and economic feasibility, although reliable indicators and data for these aspects remain limited (Fauzi et al., 2019; Chang et al., 2017; Kalvani et al., 2021; Valdivia et al., 2021).

Ethical LCA promotes responsible and transparent use of resources, supporting fair sourcing and minimizing harm to ecosystems and communities (Fauzi et al., 2019). Different approaches (consequential and attributional) reflect ethical perspectives by focusing on real-world outcomes or rule-based impact allocation, each with societal limitations (T. Schaubroeck, 2023). Ethical practice also requires stakeholder engagement, clear communication of trade-offs, and attention to social justice, human rights, and labor conditions through Social LCA and ethics audits to ensure data integrity, transparent system boundaries (Valdivia et al., 2021), and accurate interpretation is essential to prevent misleading conclusions or greenwashing (Kalvani et al., 2021).

8. Contribution to Academia and Practice

This study is expected to contribute to both academic research and industrial practice by systematically synthesizing challenges, barriers, and enablers in the implementation of Life Cycle Assessment (LCA) within manufacturing contexts. Academically, it will provide a structured typology of recurring challenges, identify methodological gaps, and propose a validated conceptual framework for integrating LCA into decision-making processes (Kiemel et al., 2022; Gómez-Garza et al., 2024; Lee et al., 2023; Mesa et al., 2021; Kumar et al., 2025). Practically, it will offer actionable guidance for industry practitioners, supporting harmonized approaches, enhancing sustainability decision-making, and informing training or knowledge transfer initiatives (Lee et al., 2023; Gulotta et al., 2023). By combining systematic literature synthesis with expert validation, the study ensures that its contributions are both theoretically rigorous and directly applicable to industrial settings, helping bridge the gap between research and real-world implementation of LCA.

9. Timeline and Deliverables

The following timeline outlines the key phases, activities, and expected deliverables for this Master's thesis, scheduled from January to early June, ensuring a systematic progression from literature review to expert validation and final reporting.

Phase	Timeframe	Key Activities	Deliverables
1. Planning & Literature Scoping	Jan – Mid Feb	Refine research questions, preliminary literature review, develop SLR protocol	Research questions, literature list, SLR protocol

2. Systematic Literature Review	Mid Feb – Mid Mar	Conduct searches, screen studies, extract and code data	Database of studies, categorized challenges/enablers, initial synthesis
3. Data Synthesis & Analysis	Mid Mar – Mid Apr	Thematic analysis, develop typologies, identify research gaps	Analytical summary, typology of challenges/recommendations, research gaps
4. Expert Validation	Mid Apr – Mid May	Conduct interviews, analyze feedback, refine framework	Validated framework, adjusted typologies, expert insights
5. Reporting & Thesis Writing	Mid May – Early Jun	Draft and finalize thesis, prepare visuals and references	Complete thesis draft, final submission-ready document

Table 2: Thesis Timeline and Key Deliverables

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